

Table S1: The 35 implemented tasks: domain, inspired paradigm, principal outputs, difficulty mechanism, software verification, and psychometric status. Exact (F) = hand-derived fixture in the frontend Vitest suite; Exact (B1–B4) = hand-derived fixture in backend pytest batch 1–4; “+ behavioural” = additional behavioural tests. NE = not established (no reliability, validity, or normative data). Levels are developer-specified; no patients with multiple sclerosis were evaluated.

Task	Domain	Inspired paradigm	Principal outputs	Difficulty / adaptation	Software verification	Psychometric status
N-Back	Working memory	Visual n -back updating [1]	accuracy, mean RT, misses, false alarms	Fixed baseline (20 trials, 1-back); adaptive version is Dual N-Back	Exact (F)	NE
Digit Span	Working memory	Digit recall (forward, backward, mixed) [2]	trial accuracy, span, score	Levels 1–10: length 3–9, direction; every fourth trial one level higher	Exact (B1)	NE
Spatial Span	Working memory	Spatial-sequence recall (Corsi-type) [3]	set and order accuracy, maximum span, score	Levels 1–10: sequence length, display and interval timing	Exact (B2)	NE
Letter–Number Sequencing	Working memory	Ordered manipulation of mixed letters and digits [2]	set accuracy, order accuracy, speed, score	Levels 1–10: sequence composition and length, timing	Exact (B2)	NE
Operation Span	Working memory	Complex span (arithmetic check plus letter recall) [4]	arithmetic accuracy, letter accuracy, dual-task score	Levels 1–10: set size, operation complexity, timing	Exact (B2)	NE
Dual N-Back	Working memory	Dual-modality (position and letter) n -back [5]	visual, auditory and dual accuracy, score	Levels 1–10: $n = 1–3$, stream length, timing, target rate	Exact (B4) + behavioural	NE
Simple Reaction Time	Processing speed	Simple reaction time [6]	mean RT, RT SD, choice accuracy, combined score	Shared processing-speed workflow; levels 1–10 change delay, alternatives, timing	Exact (F)	NE
Symbol–Digit Matching	Processing speed	Symbol–digit substitution [7]	correct, incorrect, completion time, score	Levels 1–10: symbol count, item count, exposure, target	Exact (B1)	NE
Numeric Trail Task	Processing speed	Sequential visual trail [8]	completion time, errors, score	Levels 1–10: node count, layout, expected time	Exact (B2)	NE
Pattern Comparison	Processing speed	Rapid same/different comparison [9]	accuracy, RT, timeouts, score	Levels 1–10: grid size, similarity, changed cells, duration, trials	Exact (B1)	NE
Inspection Time	Processing speed	Brief line discrimination with mask [10]	accuracy, threshold duration, score	Levels 1–10: presentation time, line difference, trials	Exact (B1)	NE
Choice Reaction Time	Processing speed	Stimulus–response choice RT [11]	accuracy, mean RT, consistency, score	Levels 1–10: 2–4 alternatives, trials, windows; up if score ≥ 83 and accuracy ≥ 82 , down if < 63	Exact (B4) + behavioural	NE
Continuous Performance Task	Attention	Sustained target detection (AX-type baseline) [12]	target hit rate, false clicks, mean RT, score	Fixed baseline (60 trials); adaptive version is Go/No-Go	Exact (F)	NE
Serial Addition	Attention	Serial addition (PASAT-type) [13]	correct, omissions, RT, accuracy-based score	Levels 1–10: shorter inter-stimulus interval, trials	Exact (B1)	NE
Color–Word Interference	Attention	Color–word interference [14]	accuracy, interference score, RT, score	Levels 1–10: color set, congruency ratio, timing, trials	Exact (B3)	NE
Go/No-Go	Attention	Response inhibition [6]	go and no-go accuracy, RT, consistency, score	Levels 1–10: stimulus similarity, no-go frequency, timing, trials	Exact (B3)	NE

Task	Domain	Inspired paradigm	Principal outputs	Difficulty / adaptation	Software verification	Psychometric status
Flanker	Attention	Flanker interference [15]	accuracy, congruent/incongruent RT, conflict control, score	Levels 1–10: congruency ratio, spacing, timing, trials	Exact (B3)	NE
Sustained-Attention Response	Attention	Frequent-go, rare-no-go vigilance [16]	overall and no-go accuracy, RT, vigilance penalty, score	Levels 1–10: stream length, target rate, timing; up if score ≥ 84 and ≤ 4 commissions, down if < 64	Exact (B4) + behavioural	NE
Task Switching	Flexibility	Cued task switching [17]	accuracy, switch cost, perseveration errors, score	Fixed baseline (40 trials); adaptive version is Rule Shift	Exact (F)	NE
Alternating Trail Task	Flexibility	Alternating number–letter trail [8]	completion time, errors, B-minus-A	Levels 1–10: node count, layout, expected time	Exact (B3)	NE
Card-Sorting Rule Shift	Flexibility	Feedback-guided card sorting [18]	categories, perseverative errors, accuracy, score	Levels 1–10: trials, rule-change frequency, feedback, category complexity	Exact (B3)	NE
Dimensional Card Sort	Flexibility	Dimensional rule switching [19]	accuracy, mean RT, switch cost, score	Levels 1–10: dimensions, trials, switch frequency, timing	Exact (B3)	NE
Rule Shift	Flexibility	Cued set shifting (color, shape, count) [17]	overall and switch accuracy, RT, perseverative errors, score	Levels 1–10: block sequence, cue time; up if score ≥ 83 and switch accuracy ≥ 72 , down if < 63	Exact (B4) + behavioural	NE
Plus–Minus Switching	Flexibility	Alternating arithmetic switching [20]	accuracy, mean RT, switch cost, score	Levels 1–10: operand range, mixed-block frequency, items, timing	Exact (B3)	NE
Tower Planning	Planning	Tower-type planning [21]	problems solved, moves, planning efficiency, time, score	Levels 1–10: problem count, disks, minimum moves, time limits	Exact (B3)	NE
Stockings-Style Planning	Planning	Tower-style planning with balls [22]	problems solved, planning efficiency, mean time, score	Levels 1–10: problems, balls, minimum moves, time limits	Exact (B3)	NE
Letter Fluency	Planning	Phonemic fluency [23]	valid unique words, errors, mean words, score	Levels 1–10: letter sets, prompt count, duration	Exact (B3)	NE
Category Fluency	Planning	Semantic fluency [23]	unique valid words, duplicates, score	Levels 1–10: category pools; expected count rises with level	Exact (B4) + behavioural	NE
Twenty Questions	Planning	Constraint-seeking problem solving [24]	identification, questions used, strategy score, score	Levels 1–10: five object pools in rotation; score multiplier $1 + 0.05(\text{level} - 1)$	Exact (B4) + behavioural	NE
Visual Search	Visual scanning	Target search among distractors [25]	targets found, accuracy, time per target, score	Levels 1–10: item count, distractor similarity, targets, grid	Exact (B4) + behavioural	NE
Cancellation	Visual scanning	Letter or symbol cancellation [26]	hits, omissions, commissions, time, score	Levels 1–10: 150–540 cells, target count and types	Exact (B4) + behavioural	NE
Feature and Conjunction Search	Visual scanning	Feature-integration search [25]	accuracy, search time, set-size slope	Levels 1–10: set size, similarity, conjunction load	Exact (B3)	NE
Landmark Line Judgment	Visual scanning	Line-bisection judgment [27]	accuracy, omissions, spatial bias, RT, score	Levels 1–10: offset distribution, exposure; up if score ≥ 84 and $ \text{bias} \leq 8$, down if < 64	Exact (B4) + behavioural	NE

Task	Domain	Inspired paradigm	Principal outputs	Difficulty / adaptation	Software verification	Psychometric status
Multiple-Object Tracking	Visual scanning	Dynamic object tracking [28]	accuracy, precision, recall, F1, score	Levels 1–10: 6–16 objects, 1–3 targets, speed, duration	Exact (B4) + behavioural	NE
Central–Peripheral Attention	Visual scanning	Divided central/peripheral attention [29]	central accuracy, peripheral localization, RT, score	Levels 1–10: presentation time, eccentricity, distractors	Exact (B2)	NE

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